



Complete Solutions – Probability

Union of Events, Conditional Probability,
Multiplication Rule, Bayes' Theorem

STA201 – Theory of Probability and Statistics

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Complete Solutions

Question 1

Given A = surface defect, B = dimensional defect. $P(A) = 50/200 = 0.25$, $P(B) = 40/200 = 0.20$, $P(A \cap B) = 15/200 = 0.075$.

- (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.25 + 0.20 - 0.075 = 0.375$.
- (b) $P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.375 = 0.625$.
- (c) $P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.075}{0.20} = 0.375$.
- (d) Check independence: $P(A)P(B) = 0.25 \times 0.20 = 0.05 \neq 0.075 = P(A \cap B)$. Not independent.

Question 2

Total = 120, $P(\text{Python}) = 70/120 = 7/12$, $P(\text{MATLAB}) = 55/120 = 11/24$, $P(\text{both}) = 30/120 = 1/4$.

- (a) $P(\text{at least one}) = 7/12 + 11/24 - 1/4 = 19/24 \approx 0.7917$.
- (b) $P(\text{neither}) = 1 - 19/24 = 5/24 \approx 0.2083$.
- (c) $P(\text{Python} | \text{MATLAB}) = \frac{1/4}{11/24} = \frac{6}{11} \approx 0.5455$.
- (d) $P(\text{MATLAB} | \text{not Python}) = \frac{5/24}{5/12} = 1/2 = 0.5$.

Question 3

Series system with $p_1 = 0.9, p_2 = 0.8, p_3 = 0.7$.

- (a) $P(\text{system works}) = 0.9 \times 0.8 \times 0.7 = 0.504$.
- (b) $P(\text{at least one fails}) = 1 - 0.504 = 0.496$.

Question 4

(Bayes' Theorem) Given: $P(D) = 0.02$, $P(+ | D) = 0.95$, $P(+ | D') = 0.10$.

$$P(+) = P(+ | D)P(D) + P(+ | D')P(D') = 0.95(0.02) + 0.10(0.98) = 0.117.$$

$$P(D | +) = \frac{P(+ | D)P(D)}{P(+)} = \frac{0.95(0.02)}{0.117} = \frac{0.019}{0.117} \approx 0.1624.$$

Bayes' explanation: We start with prior $P(D) = 0.02$. The likelihoods are $P(+ | D) = 0.95$ and $P(+ | D') = 0.10$. The denominator $P(+)$ is the total probability of a positive test. Bayes' theorem reverses the conditional: from “positive given disease” we get “disease given positive”.

Question 5

Two dice. A : sum 8 (5 outcomes), B : first die even (18 outcomes).

$$P(A) = 5/36, \quad P(B) = 18/36 = 1/2, \quad P(A \cap B) = 3/36 = 1/12.$$

$$P(A \cup B) = 5/36 + 18/36 - 3/36 = 20/36 = 5/9.$$

$$P(A | B) = \frac{1/12}{1/2} = 1/6.$$

Independence: $P(A)P(B) = 5/36 \times 1/2 = 5/72 \neq 1/12$. Not independent.

Question 6

Deck of 52. E : face card, F : heart.

$$P(E) = 12/52 = 3/13, \quad P(F) = 13/52 = 1/4, \quad P(E \cap F) = 3/52.$$

$$P(E \cup F) = 12/52 + 13/52 - 3/52 = 22/52 = 11/26.$$

$$P(E | F) = \frac{3/52}{13/52} = 3/13 = P(E).$$

Since $P(E | F) = P(E)$, E and F are independent.

Question 7

Lot 30, 6 defective, 24 non-defective. Draw 3 without replacement.

$$P(\text{all non-defective}) = \frac{24}{30} \times \frac{23}{29} \times \frac{22}{28} = \frac{12144}{24360} \approx 0.4985.$$

Question 8

(Bayes' Theorem) Let E = engineer, M = manager.

$$P(E) = 0.6, \quad P(E') = 0.4, \quad P(M | E) = 0.20, \quad P(M | E') = 0.30.$$

$$P(M) = 0.6(0.20) + 0.4(0.30) = 0.12 + 0.12 = 0.24.$$

$$P(E | M) = \frac{P(M | E)P(E)}{P(M)} = \frac{0.12}{0.24} = 0.5.$$

Bayes' explanation: Prior $P(E) = 0.6$. The evidence is “is a manager”. We update the prior using the likelihoods $P(M|E)$ and $P(M|E')$. The posterior $P(E|M) = 0.5$ reflects the probability of being an engineer after observing manager status.

Question 9

(Bayes' Theorem) Machines M_1, M_2, M_3 with priors 0.25, 0.35, 0.40 and likelihoods $P(D|M_i) = 0.01, 0.02, 0.03$.

$$P(D) = 0.25(0.01) + 0.35(0.02) + 0.40(0.03) = 0.0215.$$

$$P(M_2 | D) = \frac{0.35(0.02)}{0.0215} = \frac{0.007}{0.0215} \approx 0.3256.$$

Bayes' explanation: The question asks for the posterior probability that a defective item came from M_2 . We combine prior production share with defect rates through Bayes' theorem.

Question 10

Biased coin $P(H) = 0.6, P(T) = 0.4$. Three tosses.

- (a) all heads: $0.6^3 = 0.216$
- (b) all tails: $0.4^3 = 0.064$
- (c) at least one head: $1 - 0.064 = 0.936$

Question 11

(Bayes' Theorem) Rain R with $P(R) = 0.25$. Accident A .

$$P(A | R) = 0.08, \quad P(A | R') = 0.03.$$

$$P(A) = 0.08(0.25) + 0.03(0.75) = 0.0425.$$

$$P(R | A) = \frac{0.08(0.25)}{0.0425} = \frac{0.02}{0.0425} \approx 0.4706.$$

Bayes' explanation: Prior $P(R) = 0.25$ is updated using likelihoods of an accident under rain and no rain. The posterior $P(R|A)$ is the probability it rained given an accident.

Question 12

$P(B) = 0.4$, $P(O) = 0.3$, independent.

$$P(\text{late}) = P(B \cup O) = 0.4 + 0.3 - 0.4 \times 0.3 = 0.58.$$

$$P(\text{not late}) = 1 - 0.58 = 0.42.$$

Question 13

Total 12 balls (5R,4B,3G). Draw two with replacement.

$$(a) \quad P(RR) = (5/12)^2 = 25/144$$

$$(b) \quad P(RB) + P(BR) = 2(5/12)(4/12) = 40/144 = 5/18$$

$$(c) \quad P(RR) + P(BB) + P(GG) = 25/144 + 16/144 + 9/144 = 50/144 = 25/72$$

Question 14

$P(H) = 80/200 = 0.4$, $P(C) = 60/200 = 0.3$, $P(H \cap C) = 30/200 = 0.15$.

$$(a) \quad P(H \cup C) = 0.4 + 0.3 - 0.15 = 0.55$$

$$(b) \quad P(H' \cap C') = 1 - 0.55 = 0.45$$

$$(c) \quad P(H | C) = 0.15/0.30 = 0.5$$

$$(d) \quad P(C | H) = 0.15/0.40 = 0.375$$

$$(e) \quad P(H)P(C) = 0.12 \neq 0.15. \text{ Not independent.}$$

Question 15

(Bayes' Theorem) Suppliers A, B, C : priors 0.45, 0.35, 0.20. Defect rates 0.02, 0.03, 0.01.

$$P(D) = 0.45(0.02) + 0.35(0.03) + 0.20(0.01) = 0.0215.$$

$$P(B | D) = \frac{0.35(0.03)}{0.0215} = \frac{0.0105}{0.0215} \approx 0.4884.$$

Bayes' explanation: Even though B supplies only 35%, its higher defect rate makes it the likely source for nearly half of all defective chips.

Question 16

(Bayes' Theorem) Know K : $P(K) = 0.7$, $P(C|K) = 1$, $P(C|K') = 0.25$.

$$P(C) = 0.7(1) + 0.3(0.25) = 0.775.$$

$$P(K | C) = \frac{0.7(1)}{0.775} = \frac{0.7}{0.775} \approx 0.9032.$$

Bayes' explanation: The evidence is a correct answer. We update the prior 0.7 to a much higher posterior because knowing the answer ensures correctness, while guessing is less reliable.

Question 17

Parallel branches with $p_1 = 0.9$, $p_2 = 0.8$.

$$P(\text{works}) = 1 - (0.1)(0.2) = 0.98.$$

Question 18

10 white, 6 black. Draw two without replacement.

- (a) $P(BB) = \frac{6}{16} \times \frac{5}{15} = \frac{1}{8}$
- (b) $P(WB) = \frac{10}{16} \times \frac{6}{15} = \frac{1}{4}$
- (c) $P(B_2 | W_1) = \frac{6}{15} = \frac{2}{5}$

Question 19

$P(R) = 0.2$, $P(T) = 0.15$, $P(R \cap T) = 0.05$.

- (a) $P(R \cup T) = 0.2 + 0.15 - 0.05 = 0.30$
- (b) $P(\text{neither}) = 1 - 0.30 = 0.70$
- (c) $P(R | T) = 0.05/0.15 = 1/3$
- (d) $P(R)P(T) = 0.03 \neq 0.05$, not independent

Question 20

(Bayes' Theorem) Spam S : $P(S) = 0.2$, word "offer" O .

$$P(O | S) = 0.95, \quad P(O | S') = 0.05.$$

$$P(O) = 0.95(0.2) + 0.05(0.8) = 0.23.$$

$$P(S | O) = \frac{0.95(0.2)}{0.23} = \frac{0.19}{0.23} \approx 0.8261.$$

Bayes' explanation: Despite only 20% spam prior, the word "offer" is strong evidence, raising the posterior to about 83%.

Question 21

Total students = $25 + 10 + 10 + 8 = 53$.

- (a) $P(\text{industrial}) = 25/53 \approx 0.4717$
- (b) $P(\text{civil or electrical}) = (8 + 10)/53 = 18/53 \approx 0.3396$

Question 22

Poker hand of 5 from 52.

$$P(2 \text{ aces, } 3 \text{ jacks}) = \frac{\binom{4}{2} \binom{4}{3}}{\binom{52}{5}} = \frac{6 \times 4}{2,598,960} = \frac{1}{108,290} \approx 0.000009234.$$

Question 23

Two dice: sum 7 (6 outcomes), sum 11 (2 outcomes).

$$P = 8/36 = 2/9 \approx 0.2222.$$

Question 24

Envelopes: \$100 (75), \$25 (150), \$10 (275).

- (a) Sample space = {\$10, \$25, \$100}
- (b) $P(\$10) = 275/500 = 0.55$, $P(\$25) = 150/500 = 0.30$, $P(\$100) = 75/500 = 0.15$
- (c) $P(\text{less than } 100) = P(\$10) + P(\$25) = 0.85$

Question 25

Use inclusion–exclusion.

- (a) Smokes but does not drink: $210 - 122 = 88$, $P = 88/500 = 0.176$
- (b) Eats and drinks but does not smoke: $83 - 52 = 31$, $P = 31/500 = 0.062$
- (c) Neither smokes nor eats: $500 - (210 + 216 - 97) = 171$, $P = 171/500 = 0.342$

Question 26

$$P(\text{both defective}) = \frac{5}{20} \times \frac{4}{19} = \frac{1}{19} \approx 0.0526.$$

Question 27

Bag1: 4W,3B. Bag2: 3W,5B.

$$P(B) = \frac{4}{7} \cdot \frac{5}{9} + \frac{3}{7} \cdot \frac{6}{9} = \frac{20}{63} + \frac{18}{63} = \frac{38}{63} \approx 0.6032.$$

Question 28

- (a) Probability that a convict committed armed robbery given that they pushed dope.
- (b) Probability that a convict did not push dope given that they committed armed robbery.
- (c) Probability that a convict did not commit armed robbery given that they did not push dope.

Question 29

Husband H , wife W : $P(H) = 0.21$, $P(W) = 0.28$, $P(H \cap W) = 0.15$.

- (a) $P(H \cup W) = 0.21 + 0.28 - 0.15 = 0.34$
- (b) $P(W | H) = 0.15/0.21 = 5/7 \approx 0.7143$
- (c) $P(H | W') = \frac{0.21-0.15}{1-0.28} = \frac{0.06}{0.72} = 1/12 \approx 0.0833$

Question 30

Rejection rates $r_1 = 0.10$, $r_2 = 0.08$, $r_3 = 0.12$.

- (a) Survive first, rejected second: $0.90 \times 0.08 = 0.072$
- (b) Rejected by third: $0.90 \times 0.92 \times 0.12 = 0.09936$

Question 31

$$P(\text{incorrect and sues}) = 0.3 \times 0.9 = 0.27.$$

Question 32

Fire engines independent, $P(\text{available}) = 0.96$.

- (a) Neither available: $0.04^2 = 0.0016$
- (b) At least one available: $1 - 0.0016 = 0.9984$

Question 33

Machines B_1, B_2, B_3 : 30%, 45%, 25%; defect rates 2%, 3%, 2%.

$$P(D) = 0.30(0.02) + 0.45(0.03) + 0.25(0.02) = 0.0245.$$

Question 34

(Bayes' Theorem) Plans P_1, P_2, P_3 : 30%, 20%, 50%. Defect rates $P(D|P_i) = 0.01, 0.03, 0.02$.

$$P(D) = 0.30(0.01) + 0.20(0.03) + 0.50(0.02) = 0.019.$$

Posterior probabilities:

$$P(P_1 | D) = \frac{0.003}{0.019} \approx 0.1579, \quad P(P_2 | D) = \frac{0.006}{0.019} \approx 0.3158, \quad P(P_3 | D) = \frac{0.010}{0.019} \approx 0.5263.$$

Plan 3 is most likely. **Bayes' explanation:** We compute the posterior for each plan given the defective product. The plan with the highest posterior is the most likely source.

Question 35

(Bayes' Theorem) Latex L : $P(L) = 0.75$, $P(R|L) = 0.60$, $P(R|L') = 0.30$.

$$P(R) = 0.75(0.60) + 0.25(0.30) = 0.525.$$

$$P(L | R) = \frac{0.75(0.60)}{0.525} = \frac{0.45}{0.525} = \frac{6}{7} \approx 0.8571.$$

Bayes' explanation: The evidence is "purchases a roller". We update the prior 0.75 to a higher posterior because latex buyers are more likely to buy rollers.

Question 36

$p = 0.5, n = 4.$

- (a) Exactly 3: $\binom{4}{3}(0.5)^3(0.5) = 4 \times 0.125 \times 0.5 = 0.25$
- (b) None: $(0.5)^4 = 0.0625$

Question 37

Mistake probability $p = 0.1.$

- (a) Four each mistake: $0.1^4 = 0.0001$
- (b) Jones & Clark mistake, Roberts & Williams not: $0.1^2 \times 0.9^2 = 0.0081$

Question 38

Temperatures 3, times 4, oils 3.

- (a) Total combinations $= 3 \times 4 \times 3 = 36$
- (b) Combinations per oil $= 3 \times 4 = 12$
- (c) Permutations are not an issue because the order of factor levels does not change the recipe.

Question 39

Box: 6 black, 4 green. $P(B) = 0.6, P(G) = 0.4.$ Draw 3 with replacement.

- (a) All same colour: $P(BBB) + P(GGG) = 0.6^3 + 0.4^3 = 0.216 + 0.064 = 0.28 = 7/25$
- (b) Each colour represented: $1 - 0.28 = 0.72 = 18/25$